Regularisation and early stopping

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Keep objective changes and duration decisions distinct.

Show full gradients and miss counts,explicit indexing and unobserved future;require actual model snapshots for restoration claims.

Keep one habit: Assess independent terms and stopping/selected epochs with checked keys,keeping loss logs separate from claimed training runs.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Separate a **weight penalty** from **prediction** error

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Restore the parameters saved at the chosen checkpoint. The last parameters produced by **training** may have a different validation loss.

Opening retrieval

Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

Look for: Compute comparable **training** and validation losses at saved checkpoints

Reference example, not the only acceptable wording: Teacher explanation: **Prediction**s [3, 4], residuals [-1.5, -2], squares [2.25, 4], MSE 3.125.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an example a batch an update and an epoch

Reference example, not the only acceptable wording: Teacher explanation: Batch lengths 4, 4, 2; three updates per epoch, nine total updates and 30 example visits.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Look for: Estimate one partial derivative while other parameters stay fixed

Reference example, not the only acceptable wording: Teacher explanation: **Prediction** 3, residual 0, loss 0, gradient [0, 0].

Pictured end goal

By the end, I can explain **regularisation** and early stopping using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Training and validation curves

Batching and epochs

Gradients with a numerical check

For the compact alternative schedule, use two 45-minute sessions. Keep **prediction** error, penalty and total objective in separate columns. Teach early stopping as a second **mechanism** with a precise state machine on paper.

Cover future curve values until a decision is made; do not reward retrospective selection as if it were a live stopping rule.

Outcomes

Separate a **weight penalty** from **prediction** error

Trace a regularised update under a stated objective

Apply a stopping rule and **restore** the best observed checkpoint

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach penalty arithmetic and policy selection independently

> Main epochs start0;picture checkpoints start1;independent task states its own indexing.

Main objective,complete term updates and guided key

Keep h = ReLU(x), y = vh, train only v on (1, 2) and (2, 4), and freeze all hidden parameters and the zero **output** bias. **Training** MSE is L(v) = 2.5(v - 2) squared.

Define J(v) = L(v) + lambda times v squared, with lambda nonnegative. Only this one **output** weight is penalised here.

For lambda = 0.5, the penalty at v = 1.5 is 0.5 times 2.25 = 1.125. **Prediction** MSE is 0.625, so total objective J is 1.75. Keep these quantities separate in logs.

A smaller total objective does not automatically mean a smaller **prediction** error.

Among these three candidates, the smallest penalised objective is at v = 1.5, whereas the smallest **training** MSE is at v = 2. Penalising a weight changes the tradeoff.

It does not guarantee better validation performance or fair outcomes.

The data gradient is 5(v - 2). The penalty gradient is 2 times lambda times v, giving total gradient 5(v - 2) + 2 lambda v. At v = 1.5 and lambda = 0.5, these contributions are -2.5 and 1.5, so total gradient is -1.

A rate 0.1 step gives v = 1.6.

Assume finite numeric values and nonnegative penalty. At the new v, data loss is 0.4, penalty 1.28 and total 1.68.

Choose penalty strength using a declared validation **prediction** metric, not by comparing total objectives with different lambdas as if they were the same measure.

Main lambda.5, rate.1	Start v1.5	Next start v1.6
Data gradient 5(v−2)	-2.5	-2
Penalty gradient 2 lambda v	1.5	1.6
Total gradient	-1	-0.4
New vafter one update	1.6	1.64
New prediction MSE	0.4	0.324
New penalty	1.28	1.3448
New total objective	1.68	1.6688

Both gradient contributions are evaluated at the same old v. Keep the data term L=2.5(v−2)²and penalty lambda v²separate before adding them. The one-step code assumes finite numbers and nonnegative lambda; it is not a validator or a search for the best penalty. Comparing total Jvalues across different lambdas changes the measure being compared.

Use the same declared validation prediction metric for development choices. The early-stopping log later is a separate paper-policy exercise, not a training run generated by these steps.

Guided: miss counts at epochs 3, 4, 5, 6 are 1, 0, 1, 2. Stop at 6 and **restore** 4. Exit: using final parameters ignores restoration; selecting **patience** with test losses uses test data for development.

Restore the selected checkpoint, choose policy using validation, then freeze it before evaluating suitable untouched final examples.

Every main policy decision and unobserved future

Main supplied epoch · Validation MSE · Consecutive misses after evaluation · Best observed epoch · Policy action; patience 2

> **Main supplied epoch:** 0
>
> **Validation MSE:** 4
>
> **Consecutive misses after evaluation:** 0
>
> **Best observed epoch:** 0
>
> **Policy action; patience 2:** Save initial best

> **Main supplied epoch:** 1
>
> **Validation MSE:** 2
>
> **Consecutive misses after evaluation:** 0
>
> **Best observed epoch:** 1
>
> **Policy action; patience 2:** Strict improvement:save/reset

> **Main supplied epoch:** 2
>
> **Validation MSE:** 1.5

Consecutive misses after evaluation: 0

Best observed epoch: 2

Policy action; patience 2: Strict improvement:save/reset

Main supplied epoch: 3

Validation MSE: 1.6

Consecutive misses after evaluation: 1

Best observed epoch: 2

Policy action; patience 2: Wait

Main supplied epoch: 4

Validation MSE: 1.4

Consecutive misses after evaluation: 0

Best observed epoch: 4

Policy action; patience 2: Strict improvement:save/reset

Main supplied epoch: 5

Validation MSE: 1.45

Consecutive misses after evaluation: 1

Best observed epoch: 4

Policy action; patience 2: Wait

Main supplied epoch: 6

Validation MSE: 1.46

Consecutive misses after evaluation: 2

Best observed epoch: 4

Policy action; patience 2: Stop; select saved epoch 4

Main supplied epoch: 7

Validation MSE: 1.3

> **Consecutive misses after evaluation:** Not observed
>
> **Best observed epoch:** Not eligible
>
> **Policy action; patience 2:** Pre-recorded future value; live policy already stopped

Complete separate pictured exercises and scope

Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule.

Toy objective $J(a)=(a-3)^2+2a^2$; start a = 0; gradient 6a-6; rate 1/6. Separate validation losses: [5, 3, 3, 4], **patience** 1, no ties as improvement.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture has two separate exercises. Its penalized objective $J(a)=(a–3)^2+2a^2$, gradient 6a–6, start 0, rate 1/6 gives a1:prediction term 4, penalty 2, total 6 instead of9. Its independent supplied validation log[5, 3, 3, 4]uses checkpoint numbers 1–4. With strict improvement and patience 1, checkpoint 2is best, checkpoint 3is the first tie/miss, so stop 3 and select saved 2.

With patience 2, stop 4 and still select 2. Those losses are supplied, not computed from this one update. The main stopping log uses epochs 0–7 instead.

Trace the same strict tie rule under changed patience

AI ACADEMY / AI-5/16 · 02 · TRACE THE CASE

Follow the deciding steps

01 Regularised update

At 0 gradient -6; update gives a = 1.

02 Split objective

At 1 prediction term 4, penalty 2, total 6; at 0 total 9.

03 Early-stop separately

Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Regularised update	At 0 gradient -6; update gives a = 1.
2. Split objective	At 1 prediction term 4, penalty 2, total 6; at 0 total 9.
3. Early-stop separately	Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

Regularisation and early stopping are different tools; neither guarantees better performance in every dataset.

Change: Change **patience** from 1 to2 under the same tie rule.

Prediction: Stop at checkpoint 4; still **restore** checkpoint 2.

Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Regularisation	A constraint or training procedure intended to limit fitting choices and improve generalisation
Weight penalty	An added objective term that discourages specified large parameter values
Patience	The allowed run of consecutive non-improving evaluations before stopping under a stated rule
Restore	Reload the parameters saved at the selected checkpoint

Patience two with strict improvement

EPOCH	VALIDATION MSE	DECISION
0 then 1 then 2	4 then 2 then 1.5	Save each improved state
3 then 4	1.6 then 1.4	Wait once then reset
5 then 6	1.45 then 1.46	Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Patience two with strict improvement

EPOCH	VALIDATION MSE	DECISION
0 then 1 then 2	4 then 2 then 1.5	Save each improved state
3 then 4	1.6 then 1.4	Wait once then reset
5 then 6	1.45 then 1.46	Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

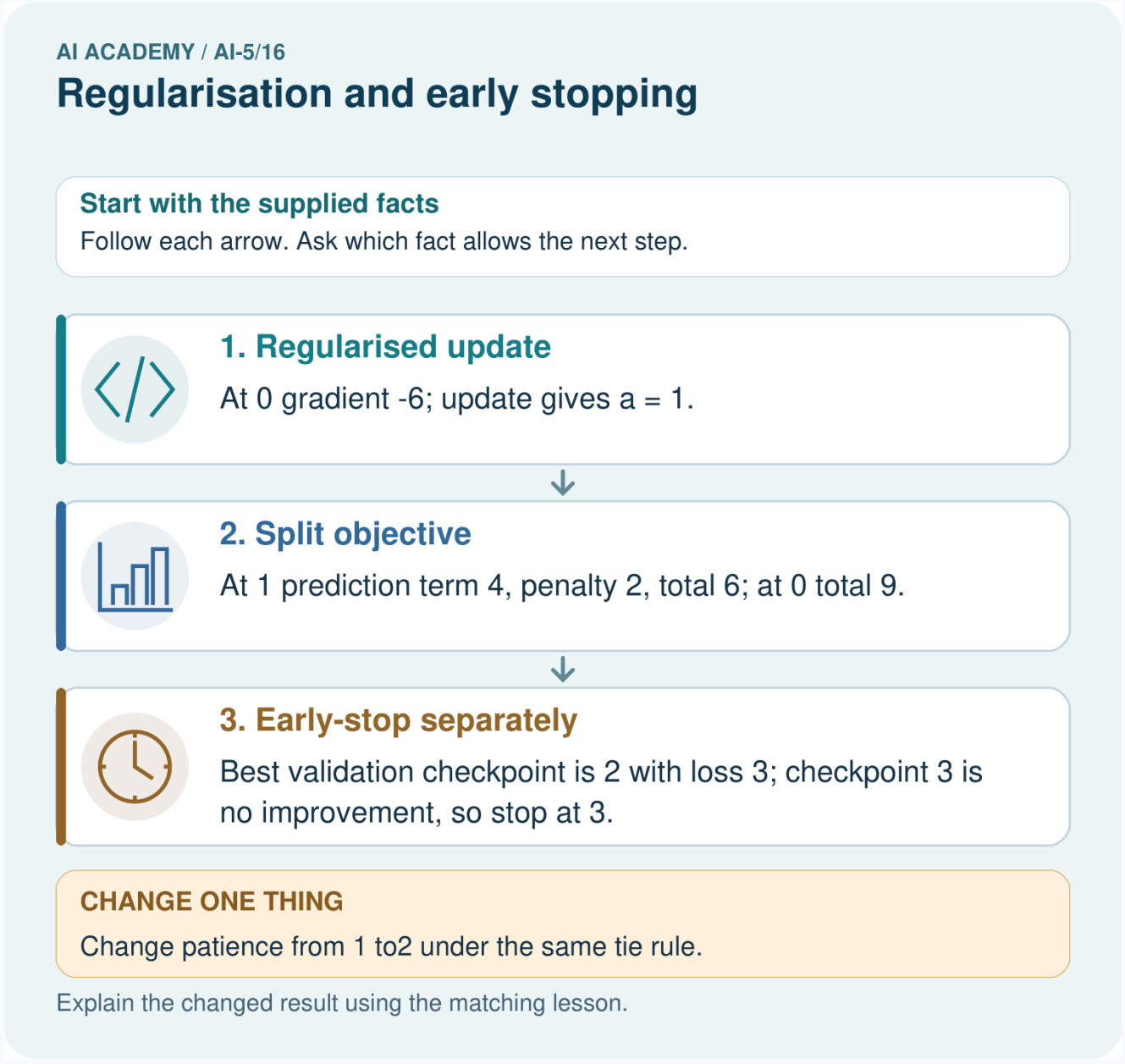
At epochs 3, 4, 5 and 6, record the miss count after evaluation. Which epoch stops training, and which epoch supplies the restored parameters?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Regularisation and early stopping

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner's reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Start v = 1.6 with lambda = 0.5 and rate 0.1. Calculate both gradient contributions, new v and new **prediction** MSE.

Teacher check: Teacher explanation: Data gradient -2, penalty gradient 1.6, total -0.4. New v = 1.64; **prediction** MSE is 2.5 times 0.36 squared = 0.324.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Regularisation** and early stopping. Proposed explanation: "Zero **prediction** error means no regularised update". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The penalty gradient can remain nonzero.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

It is part of the **training** objective, not the measured validation **prediction** loss. Report these separate quantities with their definitions. Changed-case check: Stop at checkpoint 4; still **restore** checkpoint 2.

Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Changed-case reasoning: Stop at checkpoint 4; still **restore** checkpoint 2. Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use a ledger with columns epoch, metric, best epoch and miss count; cover future rows. Extension: solve 5(v - 2) + 2 lambda v = 0 to obtain v = 10 / (5 + 2 lambda).

For nonnegative lambda, discuss why the optimum moves toward zero without claiming this always improves **prediction**.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve checkpoint restoration and introduce the named penalised weight.
7 to 18 minutes	Calculate the objective table and W1, identifying the nonzero penalty gradient at v = 2.
18 to 32 minutes	Trace the worked regularised update and use W2 for the next step.
32 to 45 minutes	Run or trace the code and distinguish objective reduction from validation evidence.
Session 2 start to 8 minutes	Reveal the validation sequence one epoch at a time; complete the guided miss counts.
8 to 20 minutes	Complete W3 before revealing the future value; compare stopping and restored epochs.
20 to 35 minutes	Collect W4 independently, scoring the arithmetic and the tied-loss policy separately.
35 to 45 minutes	Discuss W5 and the exit response, then use the fresh retry and record the complete policy.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main penalty terms and complete logs supplied. W2 new1.64/dataMSE.324. Main misses 3/4/5/6=1/0/1/2, stop 6/select 4; neverlookahead 7. W3 fullpolicy as supplied. W5 compare a fixed validation prediction metric acrosslambda, not differently defined total objectives. Independent$(a-2)^2+a^2$, gradient 4a−4, start 0, rate.25→a1, predictionterm 1, penalty 1, total 2.

Separateepochs0–4[3, 2, 2, 2, 1], patience 2:stop 3/select 1, epoch 4 unobserved. The policy program selects epochs only; actual saved model files must be loaded and verified separately.

Diagnose the current misconceptions

Zero **prediction** error means no regularised update: The penalty gradient can remain nonzero.

Early stopping adds a **weight penalty**: It limits **training** duration using a validation rule; a **weight penalty** changes the objective.

Restore the epoch where stopping happened: **Restore** the best observed checkpoint under the declared rule, which can be earlier.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: For a new objective J(a) = (a - 2) squared + a squared, gradient 4a - 4, start a = 0 and take one step at rate 0.25. Give new a, **prediction** error term, penalty and total J.

Separately apply **patience** 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and **restore**d epochs.

Teacher explanation: The update gives a = 1. **Prediction** error 1, penalty 1, total J 2. For the separate curve, epoch 1 is best at 2; ties at 2 and 3 are two misses, so stop at epoch 3 and **restore** epoch 1.

Epoch 4 is not observed by that live policy.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Separate the objective terms

At v = 2 and lambda = 0.5 in the lesson, compute **prediction** MSE, penalty, total objective and total gradient. Does zero **training** MSE make this gradient zero?

Teacher explanation: MSE 0, penalty 2, total J 2, total gradient 2. Zero data gradient does not remove the penalty gradient.

W2 Trace the next update

Start v = 1.6 with lambda = 0.5 and rate 0.1. Calculate both gradient contributions, new v and new **prediction** MSE.

Teacher explanation: Data gradient -2, penalty gradient 1.6, total -0.4. New v = 1.64; **prediction** MSE is 2.5 times 0.36 squared = 0.324.

W3 Apply patience without peeking

Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with **patience** 2. Give the stop epoch, **restore**d epoch and why the listed epoch 7 cannot change this decision.

Teacher explanation: Stop at epoch 6 after misses at 5 and 6; **restore** epoch 4 with MSE 1.4. A live run has stopped before observing epoch 7; hindsight cannot be included in that policy decision.

W5 Compare penalties fairly

A team compares lambda 0 and lambda 0.5 by choosing whichever has the smaller penalised **training** objective. Explain why this does not establish the better **prediction** model and name a suitable development comparison.

Teacher explanation: The objectives use different penalties, and both are **training** quantities. Compare the candidates on the same fixed validation examples using the same declared **prediction** metric, such as ordinary MSE.

Reserve final test examples from setting selection.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: miss counts at epochs 3, 4, 5, 6 are 1, 0, 1, 2. Stop at 6 and **restore** 4. Exit: using final parameters ignores restoration; selecting **patience** with test losses uses test data for development.

Restore the selected checkpoint, choose policy using validation, then freeze it before evaluating suitable untouched final examples.

W1 Separate the objective terms

W2 Trace the next update

W3 Apply patience without peeking

W4 Transfer both rules independently

For a new objective J(a) = (a - 2) squared + a squared, gradient 4a - 4, start a = 0 and take one step at rate 0.25. Give new a, **prediction** error term, penalty and total J.

Separately apply **patience** 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and **restore**d epochs.

Teacher explanation: The update gives a = 1. **Prediction** error 1, penalty 1, total J 2. For the separate curve, epoch 1 is best at 2; ties at 2 and 3 are two misses, so stop at epoch 3 and **restore** epoch 1.

Epoch 4 is not observed by that live policy.

W5 Compare penalties fairly

Independent assessment and fresh retry

W4 requires a = 1 and terms 1, 1, 2 plus stop epoch 3 and **restore**d epoch 1. Fresh retry: J(b) = (b - 3) squared + 0.5b squared, gradient 3b - 6; start b = 0, rate 0.25. New b = 1.5, data term 2.25, penalty 1.125, total 3.375.

On validation [5, 3, 3.2, 3.1] with **patience** 2, stop at 3 and **restore** 1.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **regularisation** and early stopping **evidence** sheet to Image-pattern Network Investigation. Show how you can separate a **weight penalty** from **prediction** error.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use a ledger with columns epoch, metric, best epoch and miss count; cover future rows. Extension: solve 5(v - 2) + 2 lambda v = 0 to obtain v = 10 / (5 + 2 lambda).

For nonnegative lambda, discuss why the optimum moves toward zero without claiming this always improves **prediction**.

Week 17 investigates initialisation and reproducibility.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/17, ai-5/19, ai-5/22. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Separate a **weight penalty** from **prediction** error

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Start v = 1.6 with lambda = 0.5 and rate 0.1. Calculate both gradient contributions, new v and new **prediction** MSE.

Reasoned practice answer: Teacher explanation: Data gradient -2, penalty gradient 1.6, total -0.4. New v = 1.64; **prediction** MSE is 2.5 times 0.36 squared = 0.324.

Check every learner before discussion: Use the worked case for **Regularisation** and early stopping. Proposed explanation: “Zero **prediction** error means no regularised update”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The penalty gradient can remain nonzero.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change **patience** from 1 to2 under the same tie rule. Require a **prediction** and a reason before comparing. Expected result: Stop at checkpoint 4; still **restore** checkpoint 2.

Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

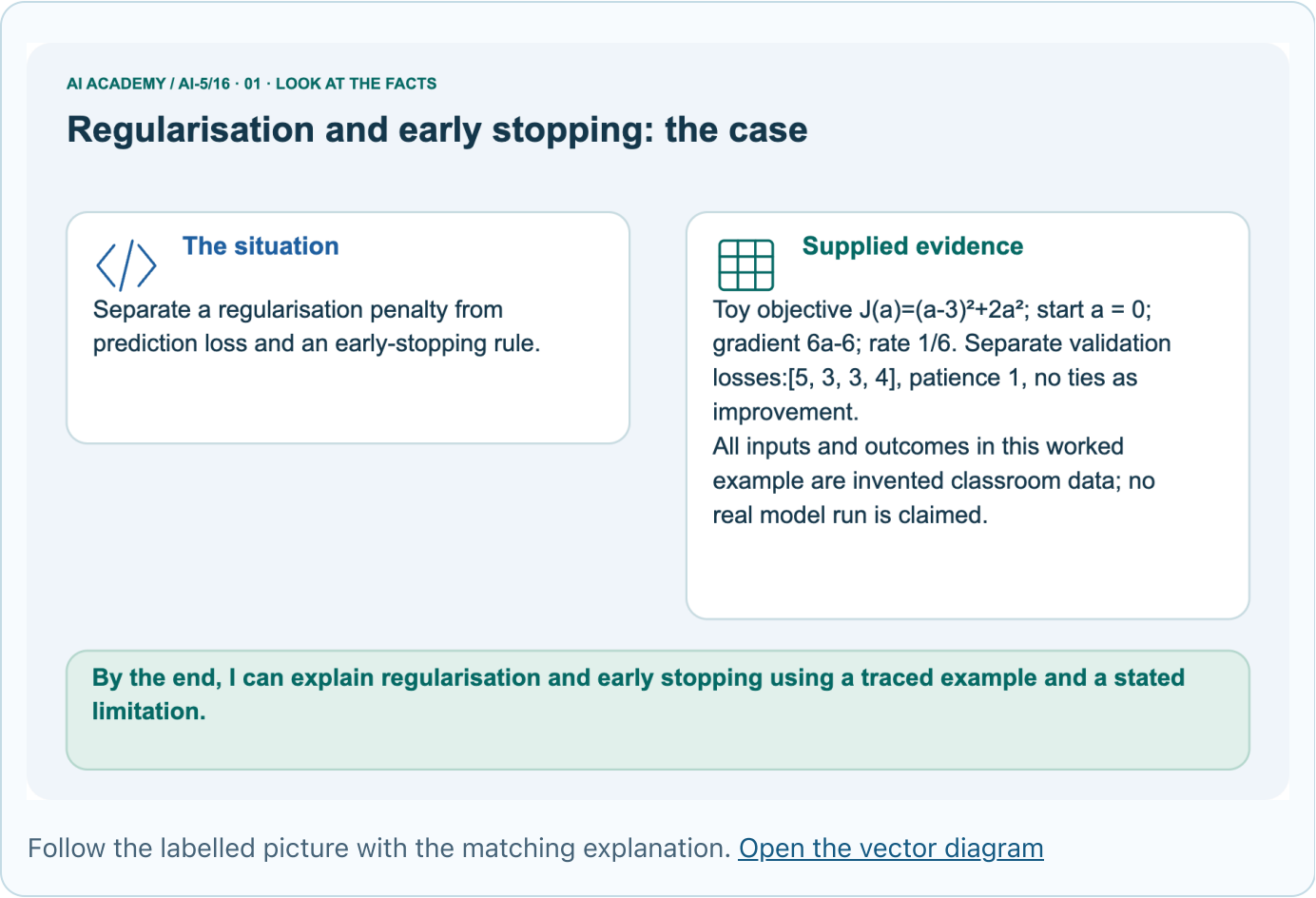
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Regularisation and early stopping: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

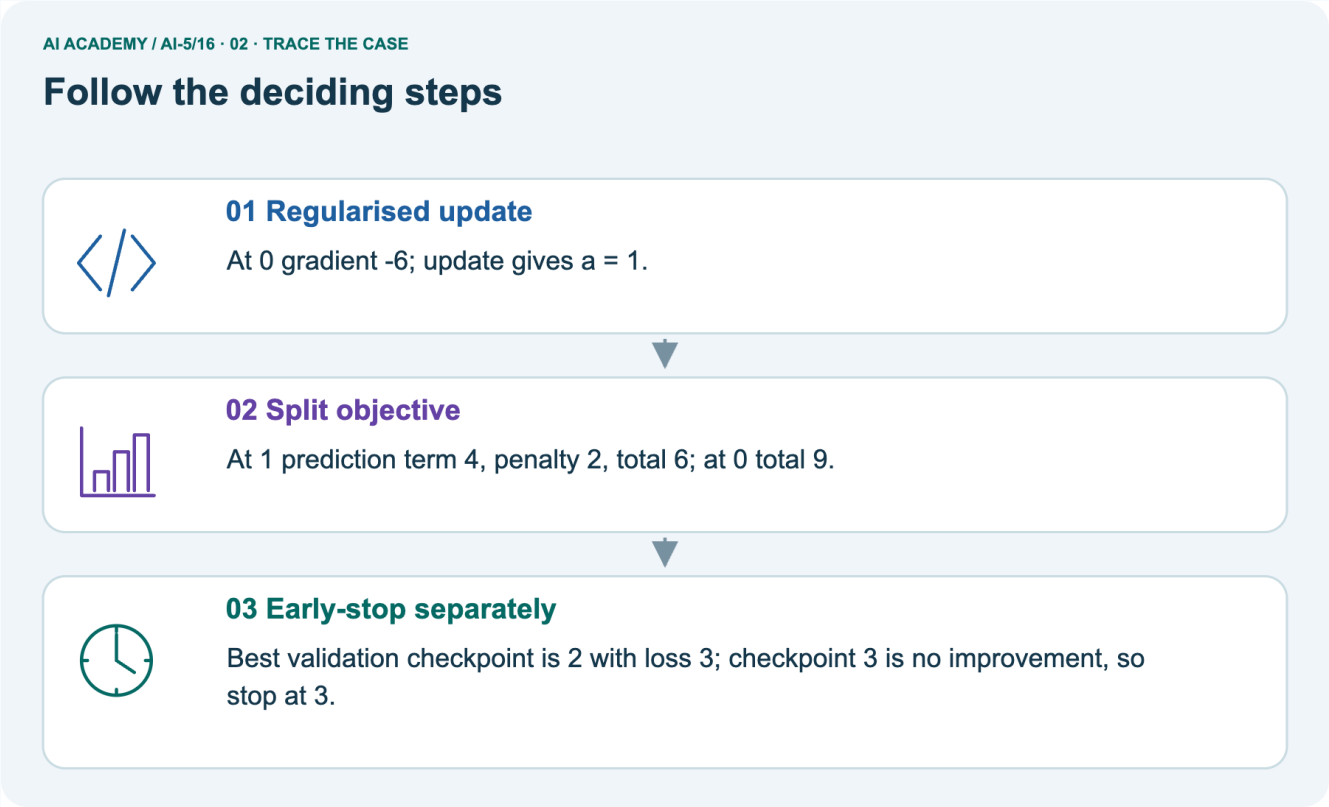
Separate a **regularisation** penalty from **prediction** loss and an early-stopping rule. Supplied facts: Toy objective J(a)=(a-3)²+2a²; start a = 0; gradient 6a-6; rate 1/6.

Separate validation losses:[5, 3, 3, 4], **patience** 1, no ties as improvement. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Regularised update: At 0 gradient -6; update gives a = 1. → Split objective: At 1 **prediction** term 4, penalty 2, total 6; at 0 total 9. → Early-stop separately: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Patience two with strict improvement



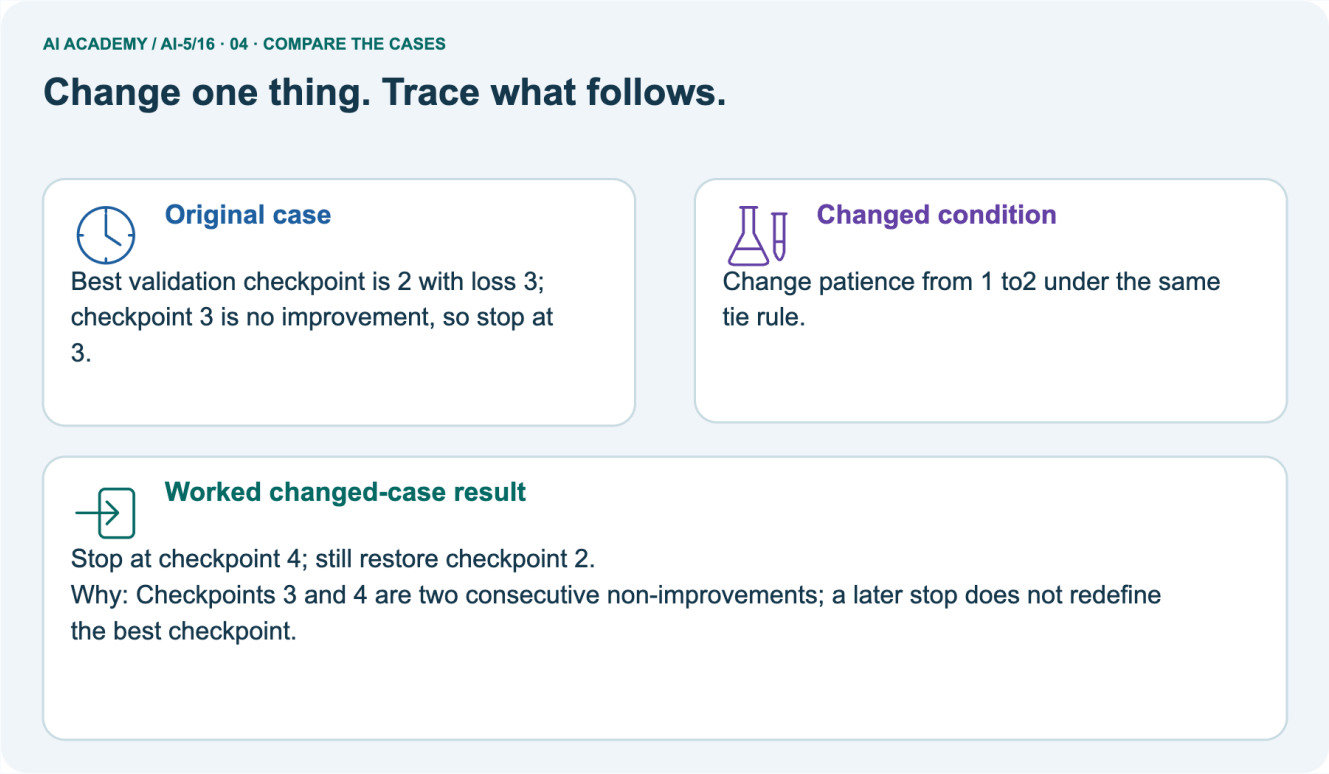
Follow the labelled picture with the matching explanation. Open the vector diagram

Patience two with strict improvement. 0 then 1 then 2 | 4 then 2 then 1.5 | Save each improved state. 3 then 4 | 1.6 then 1.4 | Wait once then reset. 5 then 6 | 1.45 then 1.46 | Two misses so stop at 6

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3. Changed condition: Change **patience** from 1 to2 under the same tie rule.

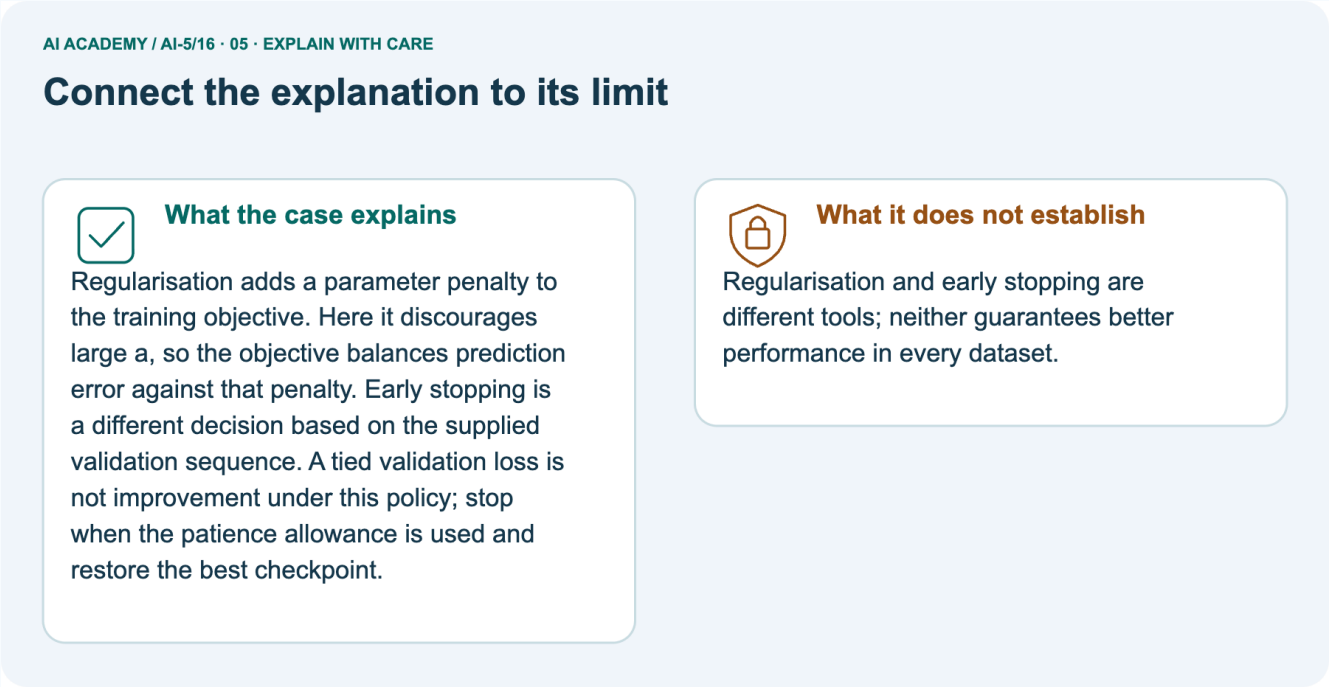
Expected result: Stop at checkpoint 4; still **restore** checkpoint 2. Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Regularisation adds a parameter penalty to the **training** objective. Here it discourages large a, so the objective balances **prediction** error against that penalty.

Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the **patience** allowance is used and **restore** the best checkpoint.

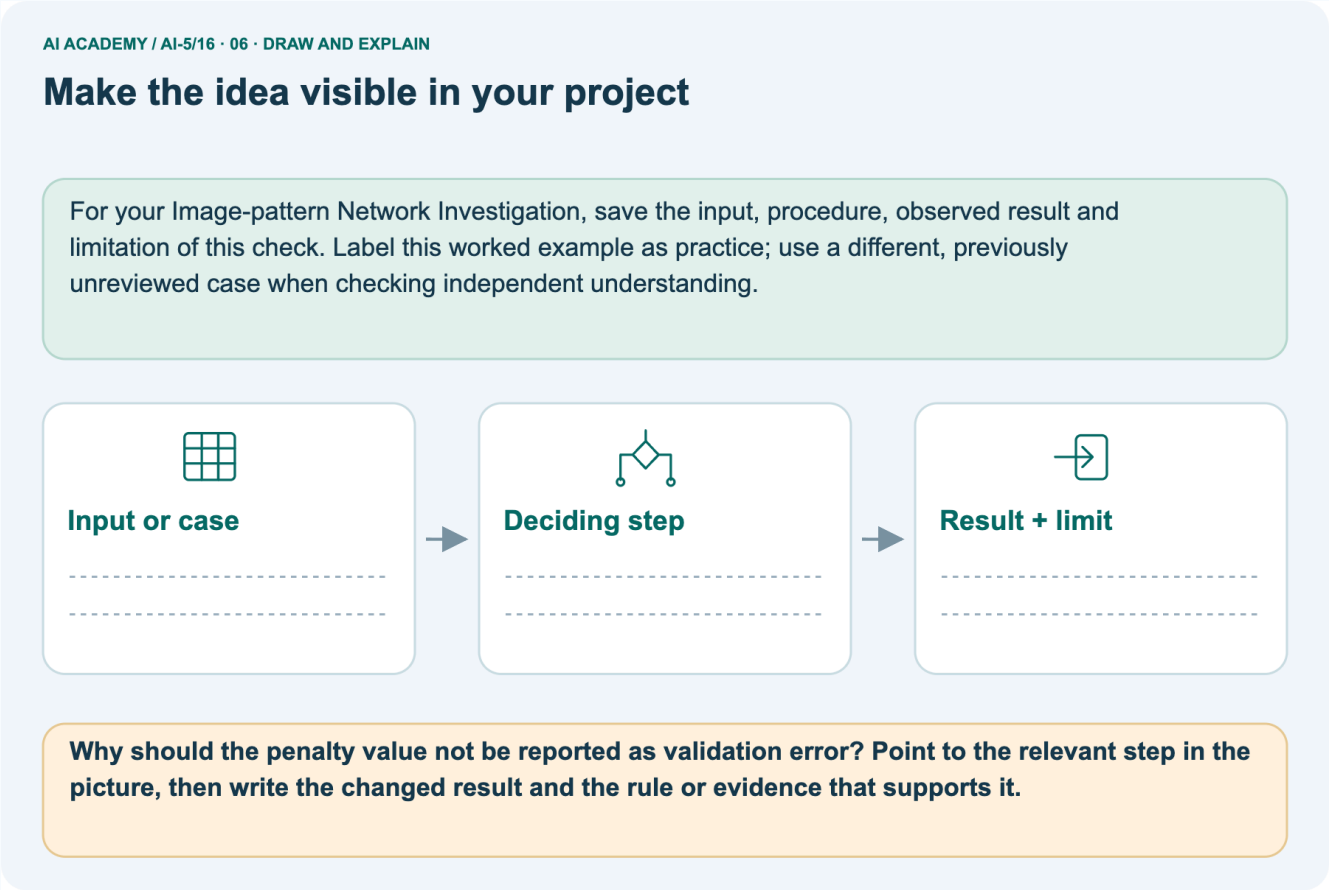
Limit: **Regularisation** and early stopping are different tools; neither guarantees better performance in every dataset.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Regularisation** and early stopping are different tools; neither guarantees better performance in every dataset.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It is part of the **training** objective, not the measured validation **prediction** loss. Report these separate quantities with their definitions.

Changed-case check: Stop at checkpoint 4; still **restore** checkpoint 2. Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Regularisation and early stopping: the case

02. Follow the deciding steps

03. Patience two with strict improvement

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.